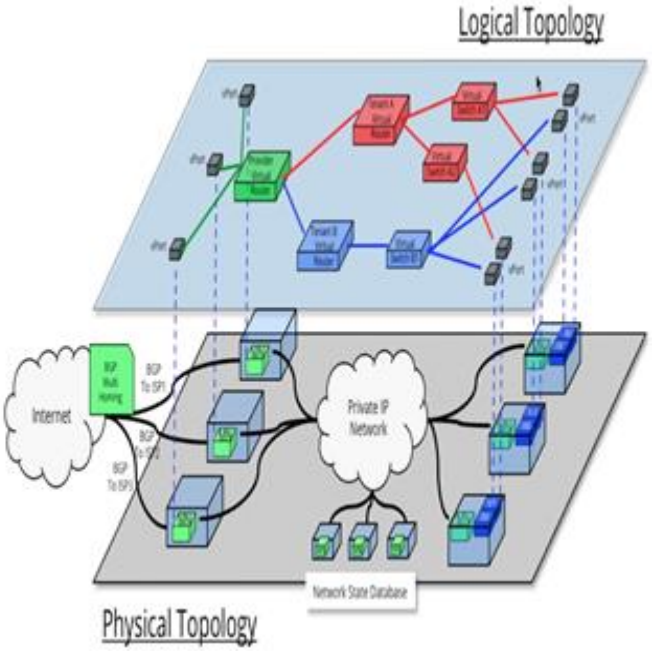


CLOUD COMPUTING COURSE

Chapter3-Network Virtualization Basics in CC.

ABSTRACT

Chapter3 Introduces physical and virtual networks in virtualization solutions.

Instructor: Asma'a Khtoom

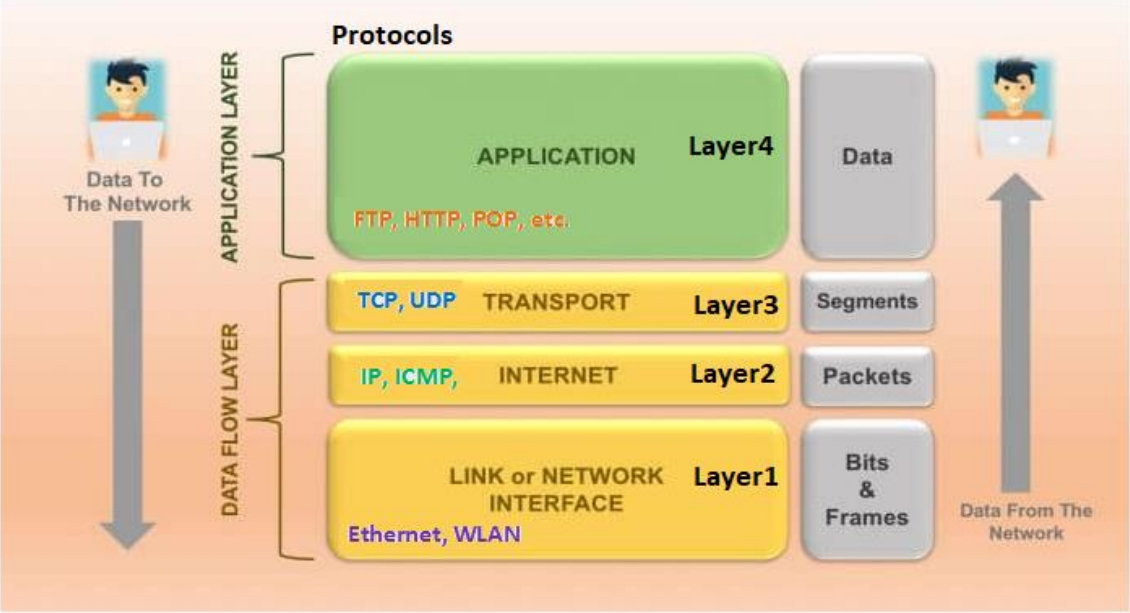
In virtualization, applications run on VMs, and VMs run on physical servers. Before connecting VMs to a network, connect physical servers to the network first. To do so, the following concepts should be discussed:

1. TCP/IP Model Suite

What is a Protocol?

A protocol is a set of rules that govern how systems communicate. For networking, they govern how **data is transferred** from one system to another.

TCP/IP protocol suite is named after two of the most common protocols – **TCP** (transmission Control Protocol) and **IP** (internet Protocol). TCP/IP protocol suite consists of many protocols that operate at one of **four layers**, as shown in the following figure.



Layer4- Application Layer: it is concerned mainly with human interaction, the implementation of software applications and related protocols. **Application layer includes** all the higher-level protocols like **DNS** (Domain Naming System), **HTTP** (Hypertext Transfer Protocol), **Telnet**, **SSH**, **FTP** (File Transfer Protocol), **RDP** (Remote Desktop Protocol) etc.

Layer3- Transport Layer: It determines how much data should be sent where and at what rate. By dividing the message received from the session layer into **segments** and numbers them to make a sequence, it contains the **port number** of the sent application. The **main protocols included** at Transport layer are **TCP** (Transmission Control Protocol) and **UDP** (User Datagram Protocol).

Layer2- Internet Layer: Internet layer pack data into data packets known as IP **Packets**, which contain **source and destination address (logical address or IP address)** information that is used to forward the packets between hosts and across networks. The Internet layer is also responsible for routing of IP packets.

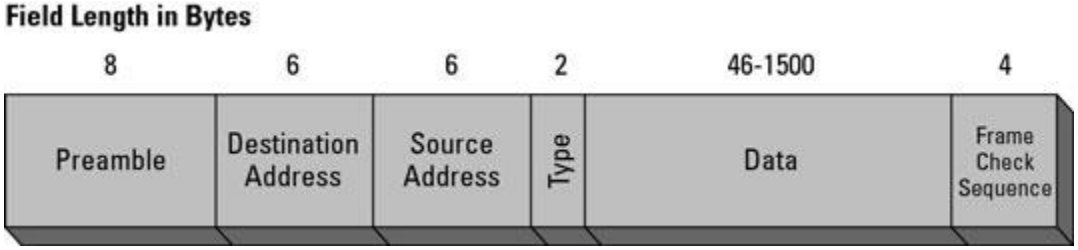
Layer1-Link Layer /Network Access: Defines details of how data is physically sent through the network, including:

- how bits are electrically or optically signaled by hardware devices that interface directly with a network medium, such as coaxial cable, optical fiber, or twisted pair copper wire.

- The MAC addresses for both **source and destination devices** are added to the data packet to create what is called Data **Frames**. The protocols **included** in this Layer are Ethernet, **Token Ring, and Frame Relay** etc.

2. Ethernet Data Frames

Data travels over the physical media of the Ethernet network in small containers, or ***frames***. There are different methods of framing Ethernet data, but the two that you are likely to see are Ethernet II and IEEE 802.3. The structure of these frames is similar, and the following explains it: As shown, the standard Ethernet frame has the following parts:



3. IP Address and Mac Address

MAC and IP address both are equally required when a device wants to communicate with another device in a network.

BASIS FOR COMPARISON	MAC	IP
Full Form	Media Access Control Address.	Internet Protocol Address.
Purpose	It identifies the physical address of a computer on the internet.	It identifies connection of a computer on the internet.
Bits	It is 48 bits (6 bytes) hexadecimal address.	IPv4 is a 32-bit (4 bytes) address, and IPv6 is a 128-bits (16 bytes) address.
Address	MAC address is assigned by the manufacturer of NIC card.	IP address is assigned by the network administrator or Internet Service Provider.

4. Physical Network Devices

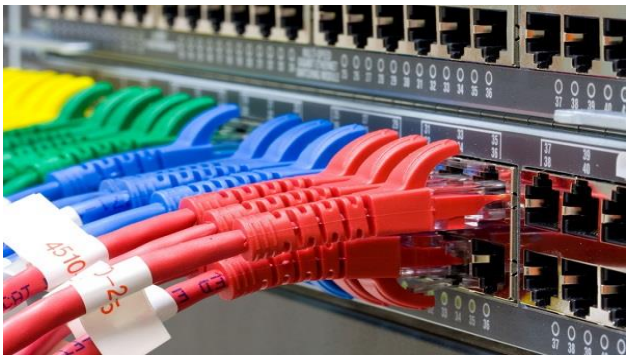
1. Router

A **router** is layer 3 hardware device that transmits data from one LAN to another if both networks support the same set of protocols. So a **router** is typically connected to at least two LANs and the **internet service provider** (ISP).



2. Switch

The Switch is a layer 2 network device that connects other devices to **Ethernet** networks through **twisted pair** cables. It uses **packet switching** technique to **receive, store** and **forward data packets** on the network.

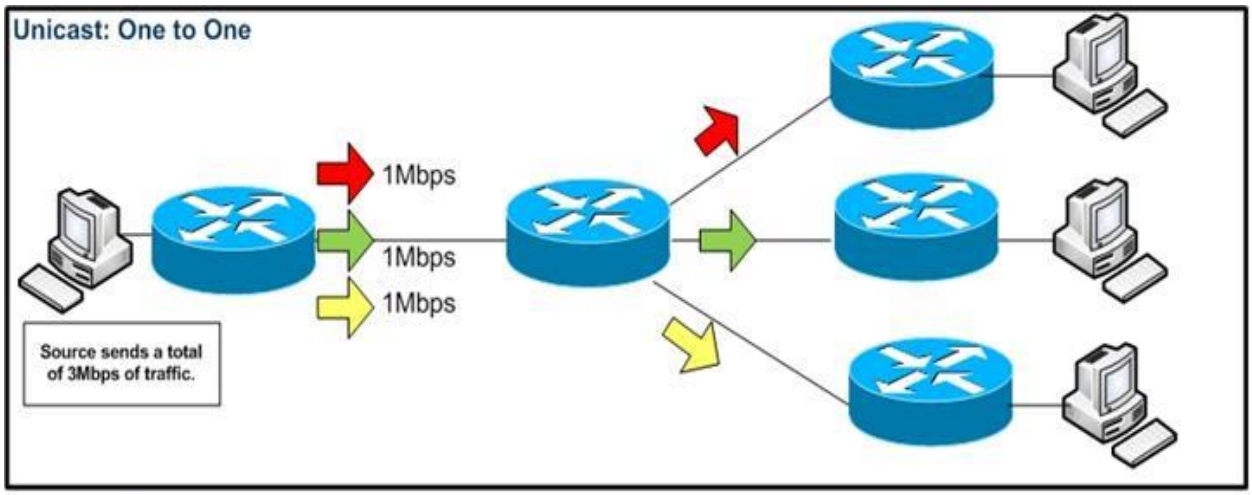


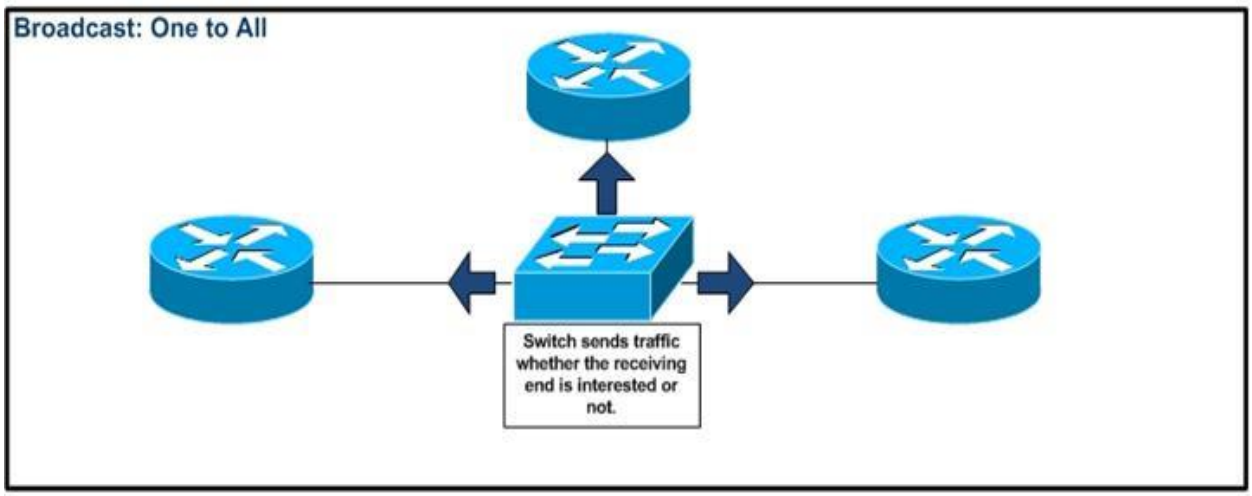
3. Hub

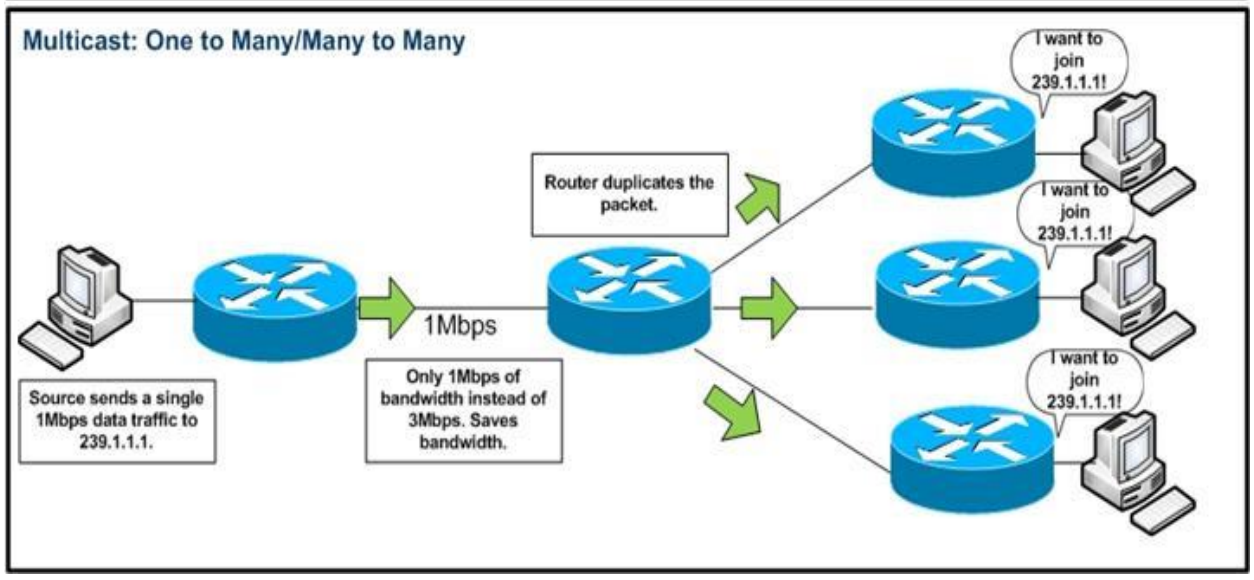
A hub is a layer1 device. The function of a hub in networking is similar to the repeater. It transfers data in the form of binary bits and uses for **broadcasting data.**



5. Network Traffic Delivery Methods







Data is transported over a network by three simple methods:

Unicast, Broadcast, and Multicast.

1. **Unicast**: from one source to one destination i.e. One-to-One

Traffic Example

Many streams of IP packets that move across networks flow from a single point, such as a website server, to a single endpoint such as a client PC. This is the most common form of information transference on networks.

2. **Broadcast:** from one source to all possible destinations i.e. One-to-All

Traffic Example

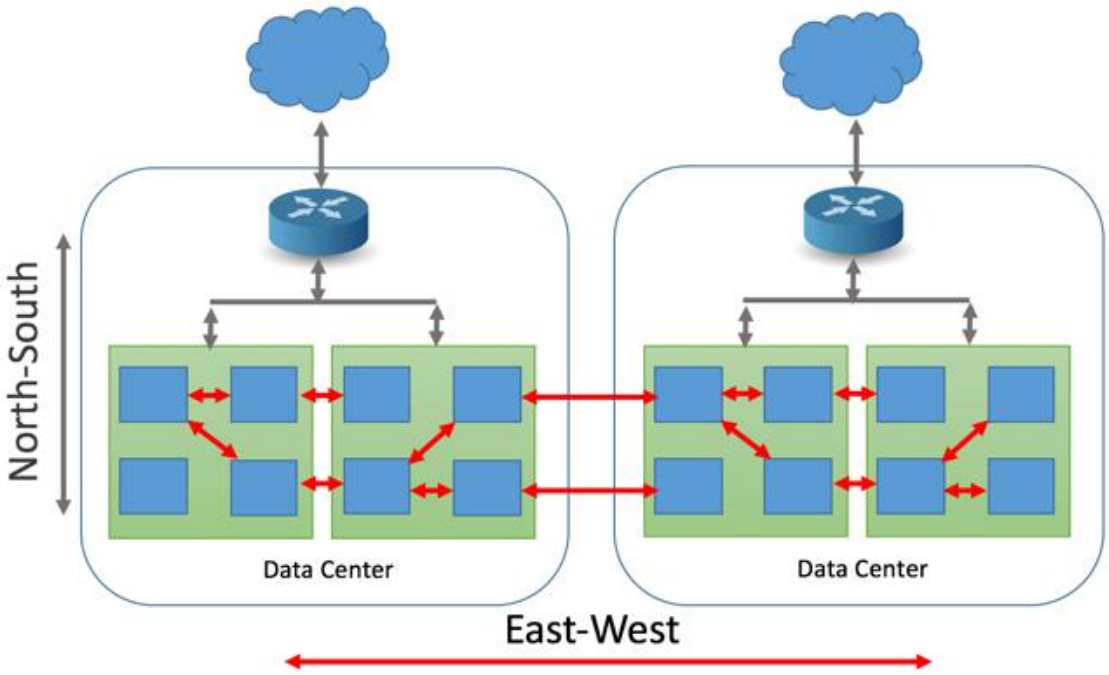
Traffic streams from a single point to all possible endpoints within reach on the network, which is generally a LAN. This is the easiest technique to ensure traffic reaches to its destinations. This mode is mainly utilized by television networks for video and audio distribution.

3. **Multicast:** from one source to multiple destinations stating an interest in receiving the traffic i.e. One-to-Many

Traffic Example

Multicast means that only the destinations that openly point to their requisite to accept the data from a specific source to receive the traffic stream. Multicast routers replicate packets received on one input interface and send the replicas out on multiple output interfaces.

6. Network Traffic Flow



NORTH-SOUTH and **EAST-WEST** traffic is a network traffic flow pattern in the context of a data center.

1. NORTH-SOUTH Traffic

The network flow between the client and the server is called the NORTH-SOUTH traffic. To put it simply, **NORTH-SOUTH** traffic is server-client traffic (**external traffic**).

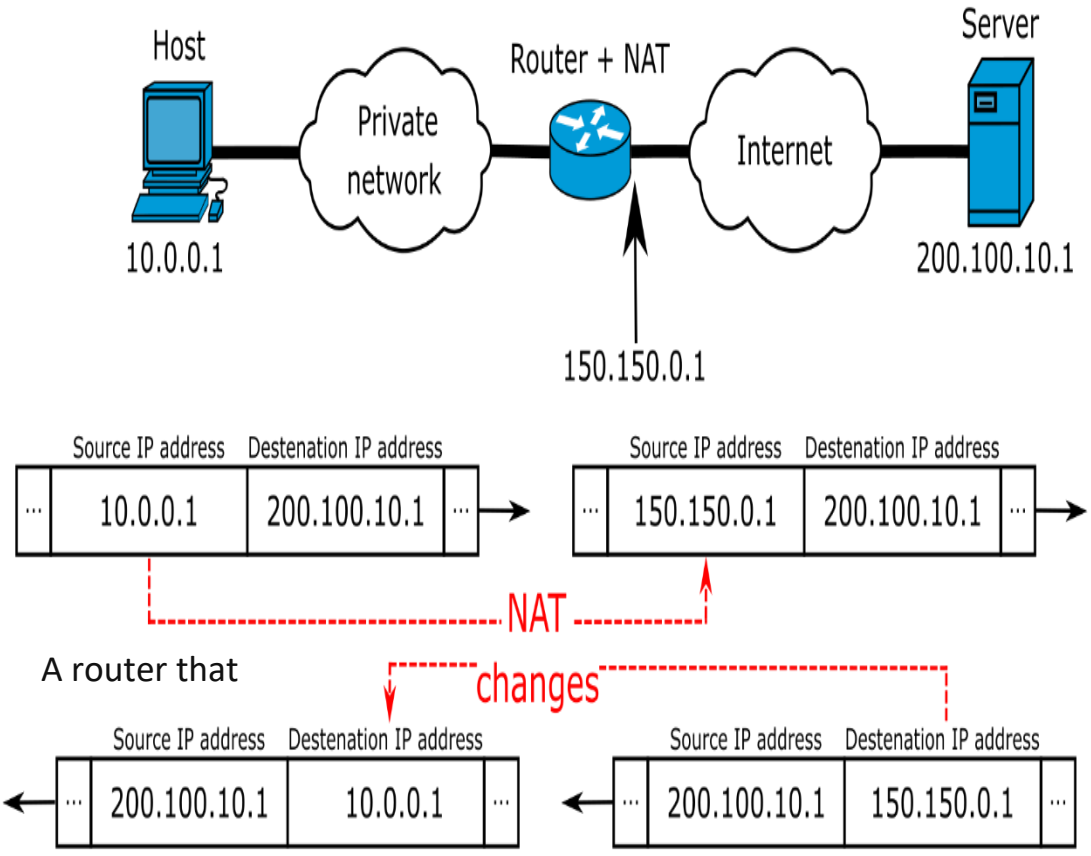
2. EAST-WEST Traffic

The network flow among different servers with in a data center (or) the network flow among different data centers themselves is called EAST-WEST traffic. To put it simply, **EAST-WEST** traffic is server-**server** traffic (**internal traffic**).

*In **Cloud Computing*** Systems, the **EAST-WEST** traffic is much more than the NORTH-SOUTH traffic.

7. NAT and IP Addresses

A NAT (Network Address Translation or Network Address Translator) is the virtualization of Internet Protocol (IP) addresses. The NAT mechanism ("natting") is a **router** feature.



Why NAT?

- Most systems using NAT do so in order to enable multiple hosts on a private network to access the Internet using a single public IP address.
- NAT helps improve security and decrease the number of public needed IP addresses.

8. Definition of NW Virtualization

Network virtualization (NV) is defined by the ability to create logical, virtual networks that are decoupled from the underlying network hardware to ensure the network can better integrate with and support increasingly virtual environments.

9. Benefit of Applying Virtualization to the Network

When applied to a network, virtualization creates a logical software-based view of the hardware and software networking resources, such as switches and routers. The physical networking devices are simply responsible for the forwarding of packets, while the virtual network (software) provides an intelligent abstraction that makes it easy to deploy and manage network services and underlying network resources. As a result, it can align the network to better support virtualized environments.

How NW Virtualization Solve Network Challenges?

Network performance is an important factor in the productivity of an organization. One of the technologies used to improve network performance is the separation of large **broadcast domains into smaller ones**. By design, routers will block broadcast traffic at an interface. However, routers normally have a limited number of LAN interfaces. A router's primary role is to move information between networks, not provide network access to end devices. The role of providing access into a LAN is normally reserved for an access layer switch.

A virtual local area network (VLAN) can be created on a **Layer 2 switch** to reduce the size of broadcast domains, similar to a Layer 3 device.

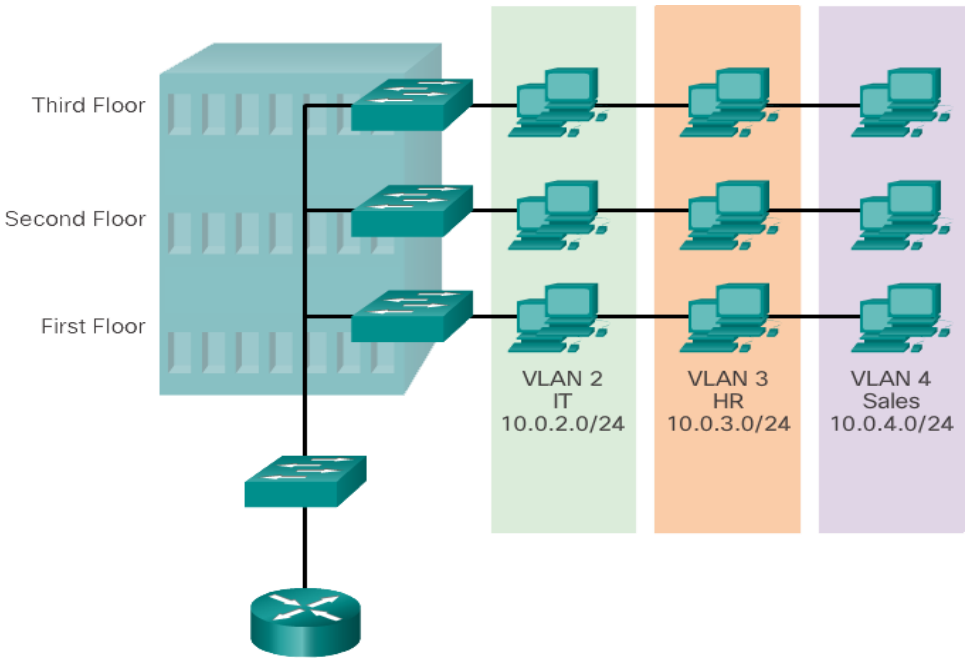
10.Virtual local area network (VLAN)

1. **VLAN Definition**: It is a logical subgroup within a local area network that is created via software rather than manually moving cables in the wiring closet. It

combines user stations and network devices into a single unit regardless of the physical LAN segment they are attached to and allows traffic to flow more efficiently within populations of mutual interest.

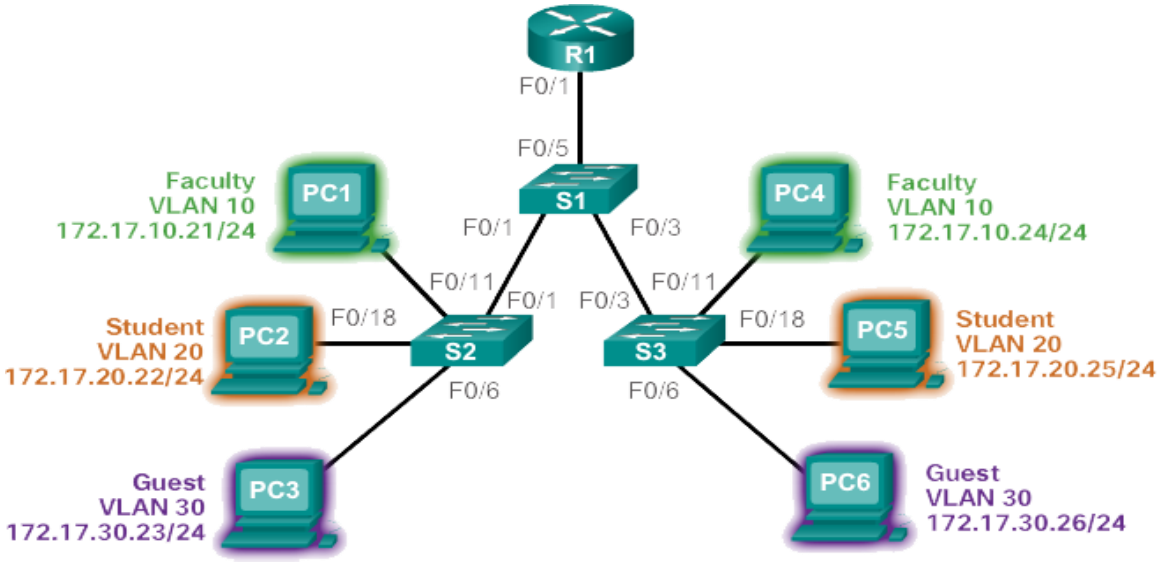
2. VLAN Concepts

- VLANs are implemented in LAN **switches (Layer 2 device).**
- VLANs reduce the time it takes to implement, moves, adds and changes.
- VLANs allow an administrator to segment networks based on factors such as function, project team, or application, **without regard for the physical** location of the user or device.



- Devices within a VLAN act as if they are in their own independent network, even if they share a common infrastructure with other VLANs.
- Any **switch port can belong to a VLAN**, and unicast, broadcast, and multicast packets are forwarded and flooded only to end stations within the VLAN where the packets are sourced.
- Each **VLAN is considered a separate logical network**, and packets destined for stations that do not belong to the VLAN must be forwarded through a device that supports routing.

- A VLAN creates a logical broadcast domain that can span multiple physical LAN segments. VLANs improve network performance **by separating large broadcast domains into smaller ones. If a device in one VLAN sends a broadcast Ethernet frame, all devices in the VLAN receive the frame, but devices in other VLANs do not.**
- **Each switch port can be assigned to only one VLAN**
- Each VLAN in a switched network corresponds to an IP network, which means that IP network numbers are applied to VLANs in an orderly fashion that takes the network as a whole into consideration.



3. VLAN Types

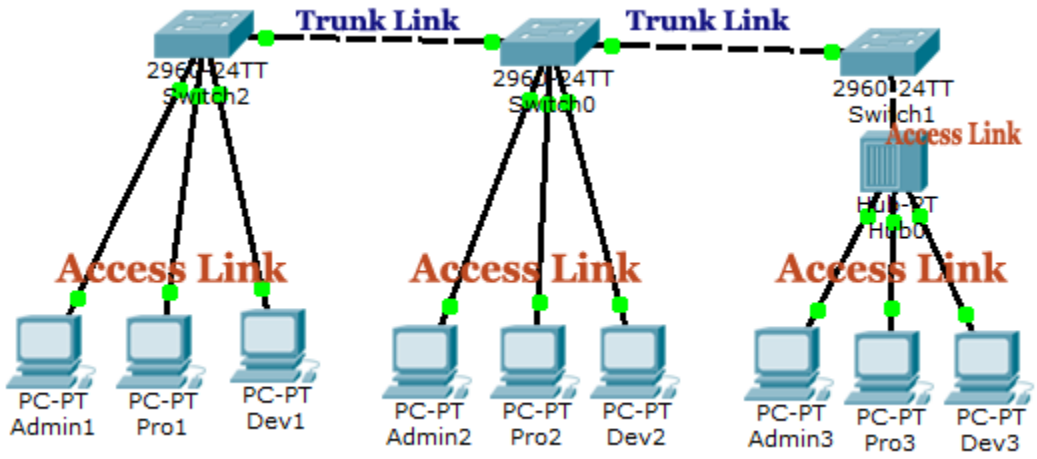
Most VLANs will be one of five main types, depending on their purpose: **Management, Data, Voice, Default and Native VLAN.**

4. VLAN Connections

During the configuration of VLAN on port, we need to know what type of connection it has. Switch supports **two** types of VLAN connection

- **Access link:** A switch port in access modes belongs to one specific VLAN and sends and receives regular Ethernet frames in untagged form. The switch interfaces connected to devices such as desktops, laptops, printers etc. are typically configured as access ports.
- **Trunk link:** Usually trunk link connection is used to connect two switches or switch to router, it carry traffic from multiple VLANs at the same time.

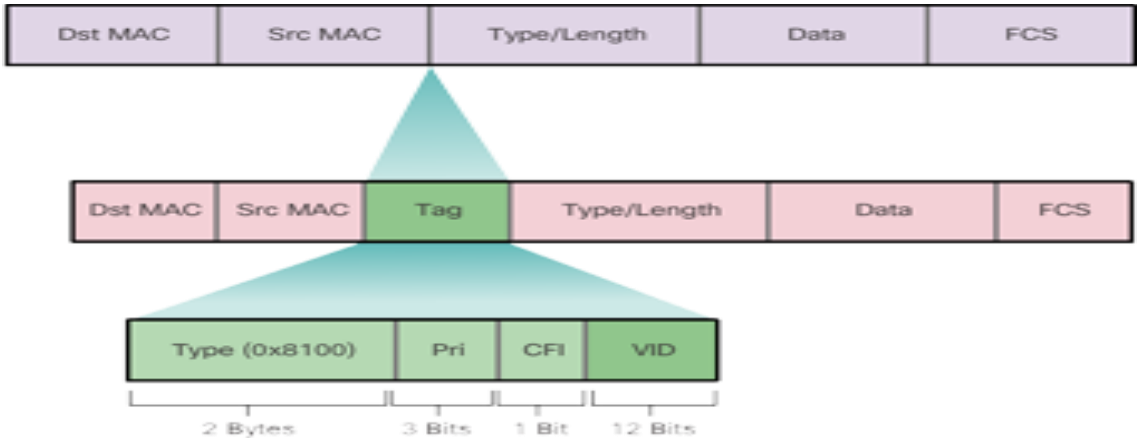
Trunking allows us to send or receive VLAN information across the network. To support trunking, original Ethernet frame **is modified** to carry VLAN information.



5. Tagging Ethernet Frames for VLAN Identification

When Ethernet frames are placed on a trunk, information about the VLANs to which they belong must be added.

This process, **called tagging,** is accomplished by using the **IEEE 802.1Q header**, specified in the IEEE 802.1Q standard. The 802.1Q header includes a **4-byte** tag inserted within the original Ethernet frame header, **specifying the VLAN to which the frame belongs**.



After the 802.1Q standard defines the VLAN frame, only certain interfaces of a device can identify a VLAN frame. Based on their ability to identify VLAN frames, interfaces are divided into two types:

- **Access interface**: The access interface resides on the switch and is used to connect the interface of the host. That is, the access interface can only connect the access link. Only packets carrying the default VLAN ID of this interface can pass through. Also, Ethernet frames sent through an access interface do not carry any tag.

- **Trunk interface:** A trunk interface is used to connect a switch to other switches and can connect only a trunk link. A trunk interface permits the tagged frames of multiple VLANs to pass through.

Interfaces of each type can be assigned with a default VLAN ID called PVID (Port Default VLAN ID). The meaning of a default VLAN varies with the interface type. The default VLAN ID of almost all switches is 1.

The following table lists the methods of processing data frames on various interfaces.

Table 3-1 Methods of processing data frames

Interface Type	Receiving Untagged Packet	Receiving Tagged Packet	Sending Frame
Access interface	Receive the packet, and tag it with the default VLAN ID.	If the VLAN ID of the packet is the same as the PVID of the interface, receive the packet. Otherwise, discard the packet.	Strip the PVID tag from the frame, and then send the frame.
Trunk interface	Tag the packet with the default VLAN ID. If the default VLAN ID is allowed on the interface, receive the packet. Tag the packet with the default VLAN ID. If the default VLAN ID is not allowed on the interface, discard the packet.	If the VLAN ID is allowed on the interface, receive the packet. If the VLAN ID is not allowed on the interface, discard the packet.	If the VLAN ID is the same as the default VLAN ID and is allowed on the interface, untag and then send the packet. If the VLAN ID is different from the default VLAN ID and is allowed on the interface, keep the tag unchanged, and send the packet.

Examples

1. In Huawei FusionCompute, the VLAN ID of a port group is set to 10, which indicates ( ).

A. **The system will untag the data frames with the VLAN 10 tag that pass through the port group and then forward the data frames.**

B. **The system will add the VLAN 10 tag to the data frames without a VLAN tag that pass through the port group and then forward the data frames.**

C. **The system will discard the data frames with a tag other than VLAN 10 that pass through the port group.**

D. The system will change the tag of data frames with a non-VLAN-10 tag that pass through the port group to VLAN 10 and then forward the data frames.

2. The port configuration of Huawei switch is as follow:
   Port link-type **trunk**
   Port trunk allow-pass vlan 16 17

   Which of the following is **true** based on the preceding configuration?
   A. Data frames with the VLAN 16 tag will be untagged and forwarded when passing through this port.
   B. Data frames with the VLAN 17 tag will be forwarded with the tag retained when passing through this port.
   C. Data frames without any tags will be discarded when passing through this port.
   D. Data frames without any tags will be tagged with VLAN 16 or VLAN 17 when passing through this port and then forwarded.

3. The port configuration of Huawei switch is as follow:
   Port link-type **access**
   Port default vlan 16

   Which of the following is **true** based on the preceding configuration?
   A. This port doesn't perform any operation on the data frames that pass through it.
   B. This port removes the VLAN 16 tag from the data frames and forward the data frames.
   C. This port adds the VLAN 16 tag to the data frames that pass through it.
   D. This port allows **only** the data frames with the VLAN 16 to pass through

6. Benefits of VLANs

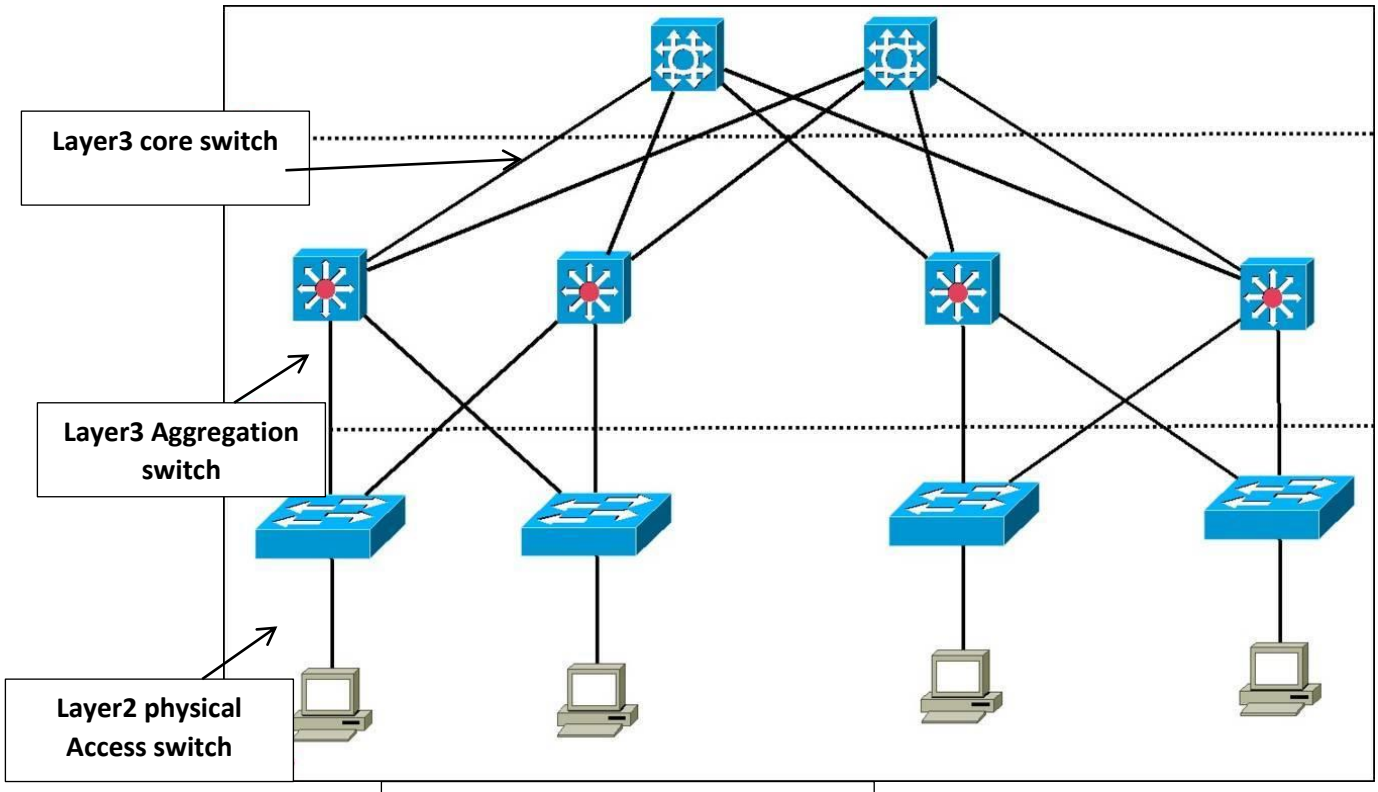
The primary benefits of using VLANs are as follows:

- **Isolate broadcast domains**
- **Enhance Security** - Groups that have sensitive data are separated from the rest of the network, decreasing the chances of confidential information breaches.
- **Improve network robustness**

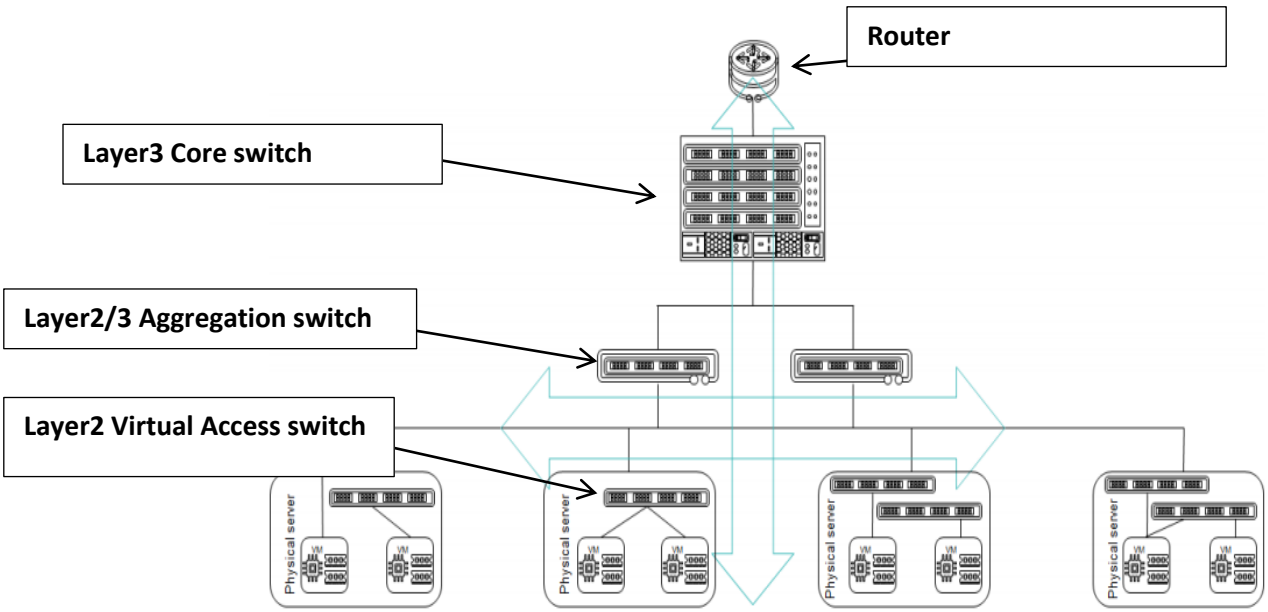
11.Physical & Virtual Network Devices in Cloud

In virtualization, applications run on VMs, and VMs run on physical servers. Before connecting VMs to a network, connect physical servers to the network first. To do so, the following devices are required:

1. Routers: it is layer3 device and it has routing function, usually deployed to connect to the internet.
2. Layer3 switches: it is layer3 device and it has routing function, it is performance is better than router and is applied to the LAN with frequent data exchange.
3. Layer2 switches: it is layer2 device and it is supports VLAN configuration only.
4. Server physical NICs/VM virtual NICs: it works at layer1
5. Virtual Layer2 switches: As cloud computing and virtualization are becoming increasingly popular, layer-2 access switches no longer function as the network access layer. Instead, layer-2 access switches will be deployed on servers to connect VMs. In this case, virtual switches are required and they function as a real network access layer.



Physical Network Architecture

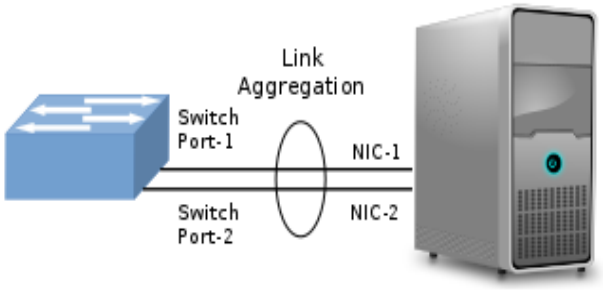


Cloud Network Architecture with Virtual Devices

A physical server uses its own physical NIC to connect to the network. All VM traffic enters the entire network through various types of network ports. The physical NIC involves a **key concept, which is port (link) aggregation**. Port aggregation indicates that **multiple physical Ethernet links are bonded into one logical link to increase the link bandwidth.** In addition, the bundled links dynamically back up each other, greatly improving link reliability.

LACP protocol allows multiple physical ports to be bonded into a logical port to **increase the link bandwidth** without upgrading hardware. In addition, the **link backup mechanism** of LACP provides higher link transmission reliability.

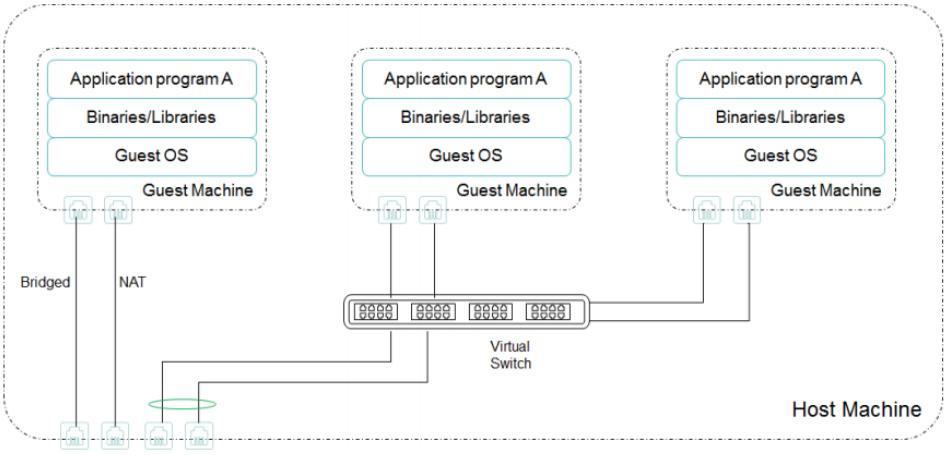
LACP Link Aggregation Control Protocol



12.Networks between VM's

Virtual Machines (VMs) run isolated from each other on one large host. However, as conventional computers communicate over a network, VMs can also be connected in different ways.

- A Virtual Machine (VM) is a combination of many virtual resources and among them **is one or more virtual network interfaces**.
- The **hypervisor** or Virtual Machine Manager (VMM), which is the software layer responsible for managing VMs, prepares these network interfaces and make them appear to the guest operating system, which runs inside the VM, as conventional Network Interface Cards (NICs) with **MAC addresses** assigned on creation.
- The VMM uses previous virtual network devices in different combinations to allow scenarios where VMs connect with each other and with the external network; the host machine has access to.
- A common virtualization system uses the architecture shown in the next figure. In a **personal or small-scale** virtualization system, VMs are **bound to physical NICs using bridges or NAT**. In a **large-scale** corporate virtualization system, VMs are connected to physical networks using **virtual switches**.



Architecture of Virtual Network

Currently, each virtualization vendor has its own virtual switching product, such as VMware vSwitch, Cisco Nexus 1000V, and **Huawei DVS**.

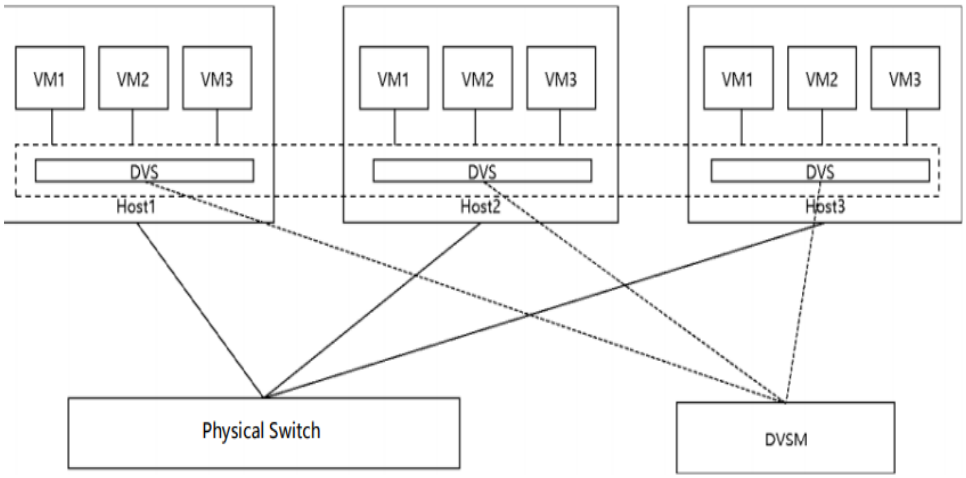
The virtual switch can be

1. A **common virtual switch**: is layer2 switch, runs only on a single physical host. All network configurations apply only to VMs on the physical host.
2. A **distributed virtual switch** (DVS): is layer2 switch, deployed on different physical hosts. You can use the virtualization management tool to configure distributed virtual switches in a unified manner. The distributed virtual switch is **required for VM live migration**. **Huawei virtualization products use distributed virtual switches.**

13.Network Features of Huawei Virtualization Products

1. Network Solutions in Huawei Virtualization Products

Huawei distributed virtual switches can be centrally managed. The centralized management modules provide a unified portal for simplified configuration management and user management.



DVS

The virtual switches distributed on physical servers provide VMs with a range of capabilities, including layer-2 connectivity, isolation, and QoS.

1. The DVS model has the following characteristics:
    a. Multiple DVSs can be configured, and each DVS can serve multiple CNA nodes in a cluster.
    b. A DVS provides several virtual switch ports (VSP) with configurable attributes, such as the rate and statistics.
    c. Different physical ports can be configured for the management plane, storage plane, and service plane.
    d. Each VM provides multiple virtual NIC (vNIC) ports, which connect to VSPs of the switch in **one-to-one** mapping.

2. **Port group and uplinks**

- Ports with the same attributes are assigned to the **same port group** for easy management. Ports that belong to the **same port group** are assigned the **same VLAN.**
- The configuration of VM port attributes can be simplified by configuring port group attributes instead, including security and QoS.
- Modifying port group attributes has no impact on the proper running of VMs.
- Port group attribute changes do not affect VM running.

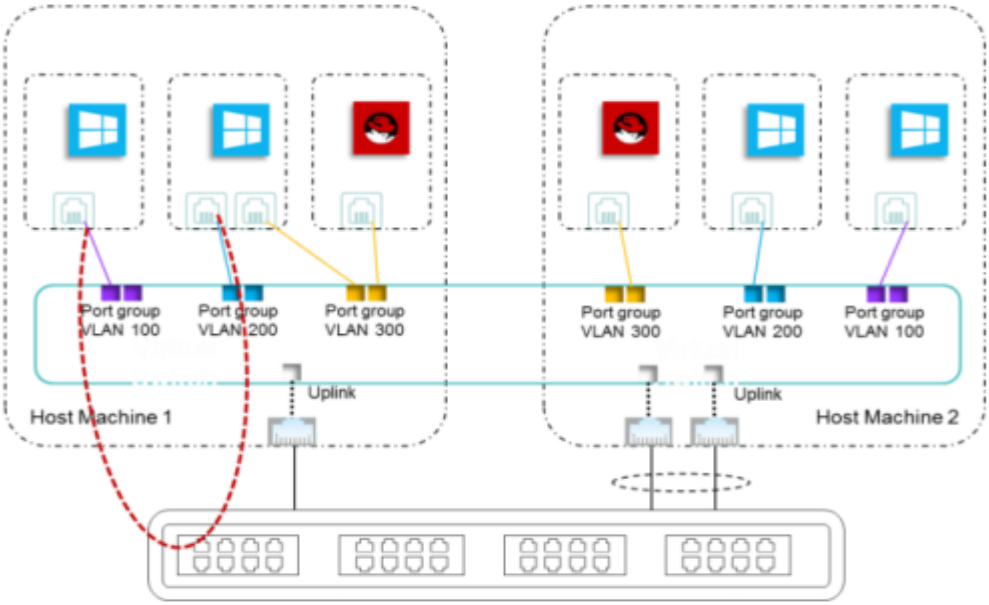
Uplink: the port that connects the host and the DVS.

- An **uplink port** or an **uplink port aggregation group** can be configured for each DVS to enable external communication of VMs served by the DVS.
- An uplink aggregation group comprises multiple physical NICs working based on preconfigured load-balancing policies.
- Administrators can query information about an uplink, including its name, rate, mode, and status.

- Uplink aggregation allows multiple physical ports on a server to be bound as one port to connect to VMs.
- Administrators can set the bound port to **load balancing mode or active/standby mode.**

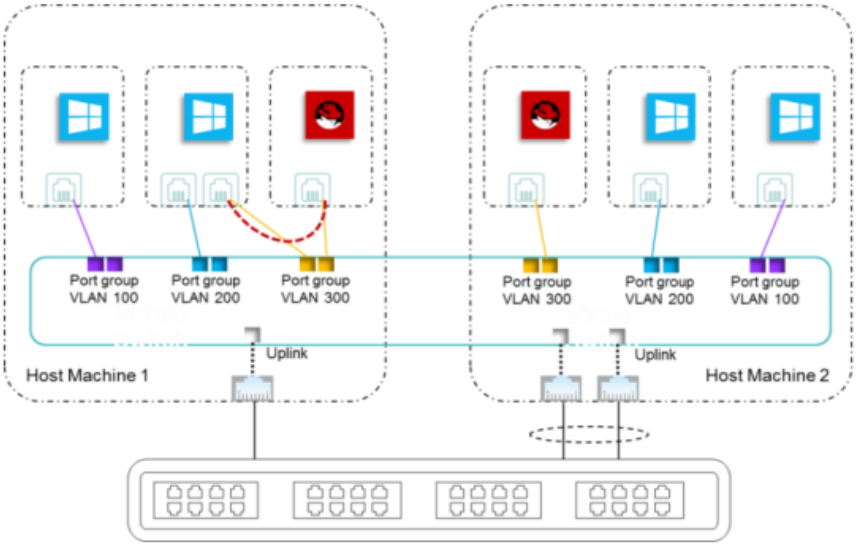
2. DVS Traffic Flow

Scenario1: **VMs run on the same host but belong to different port groups.**



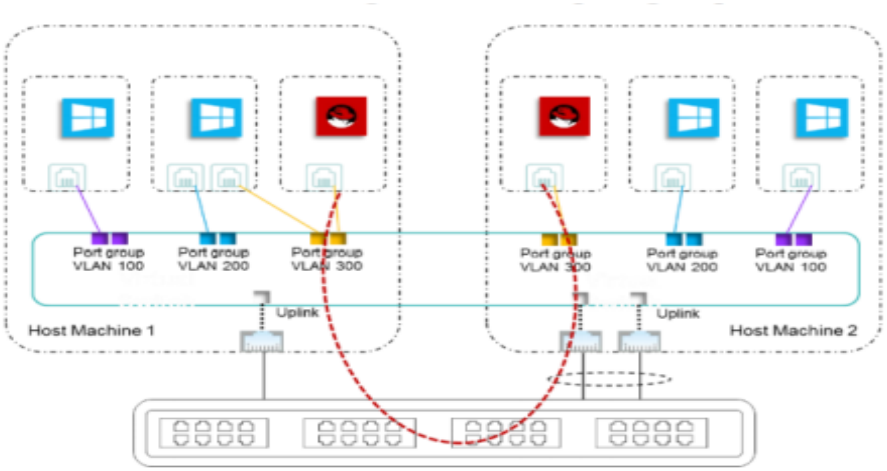
If two VMs belong to different port groups, they are associated with different VLANs so when two VMs are associated with different VLANs, they are configured with IP addresses that belong to different IP address ranges. Enabling communication between the two VMs requires a layer-3 device, which can be a layer-3 switch or router.

Scenario2: VMs run on the same host and belong to the same port group.



When two VMs run on the same host and belong to the same port group, they belong to the same broadcast domain, in which case, **they can communicate with each other through a virtual switch and the traffic will not enter** the physical network.

Scenario3: VMs run on different hosts but belong to the same port group



When two VMs that run on different hosts but belong to the same port group intend to communicate with each other, the traffic needs to pass through **the physical L2 switch**. The two VMs can **communicate with each other without using the layer-3 device**, which is different from the situation when the two VMs run on different physical servers and belong to different port groups.

Scenario4: Multiple DVSs run on a physical host.

When two VMs are connected to different DVSs, the port groups associated with the two DVSs have different VLAN IDs, which means that the two VMs use different IP addresses. In this case, the traffic between the two VMs will need to be routed through **a layer-3 device.**

3. Security Group

- A security group is **a logical group that consists of instances that have the same security protection requirements and trust each other in the same region.**

- All VM NICs in a security group communicate with each other by complying with the security group rules. A VM NIC can be added **to only one security** group.

- Users can create security groups based on VM security requirements.

- Each security group provides a set of access rules. VMs that are added to a security group are protected by the access rules of the security group.

- Users can add VMs to security groups for security isolation and access control when creating VMs.

- The security group provides a similar function **as the firewall does**.

1. Security group:

- **Security groups protect VM's based on rules not based on port groups.**
- **Security groups is a set of rules like firewall**
- **One physical NIC can be added only to one security group.**
- **One VM can be protected by a single security group**

2. Port group

- **Port group is an aggregation of virtual ports on DVS.**
- **VM NIC is connected to a virtual port in a port group.**
- **One DVS can have multiple port groups**
- **Port group consist of multiple ports with the same attributes**
- **Port group attribute change, not affect VM running.**

3. The uplink

- **Uplink is link between DVS and physical NIC of the host.**
- **One host can has multiple uplinks connected to different DVS.**
- **Multiple DVS cannot share an uplink.**
- **Uplink can correspond to one physical network port or logical port that binds multiple physical network ports.**